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A Blueprint for a Successful Tight Reservoir Field Development and Reservoir Management

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Abstract

This paper is a study case of the field development of a tight carbonate reservoir. Development challenges are summarized, and the effective technologies used to solve them are listed. The combination of the field development steps taken and the technologies deployed can be used as a Blueprint to develop and manage analogue reservoirs.

Introduction

The subject field (referred to as Field-F throughout this paper) is a tight reservoir with a permeability ranging from 0.01 to 3 mD. This low permeability poses a serious challenge in Field Development and Reservoir Management but considering the size of the prize, a STOIP in the hundreds of mln bbls, it is particularly important to deploy technologies and find solutions that would allow us to unlock such resources. Since the beginning of Field-F development, a few technologies have been successfully deployed that shaped the development and reservoir management of Field-F as a Blueprint for the successful development and management of analogue reservoirs.

The first issue that has been successfully addressed was the well productivity. Two technologies have been successfully deployed in Field-F, Hydraulic Fracking and Fishbone Stimulation. In both cases and despite the operational challenges, productivity enhancement has been significant and sustainable. Sustained production from the Hydraulically Fracked well has increased five-fold while from the Fishbone Stimulated well, three-fold. Pending approval, a Fishbone completion could be deployed in one additional well, further consolidating productivity gains. Finally, the option of Advanced Stimulation for the producers is considered, and it will be tested in selected producers. This will increase the well productivity at an optimized cost.

The second issue that has been successfully addressed was the limited water injectivity. A successful Advanced Stimulation Campaign has been completed in 2023. This stimulation campaign has achieved a sustained injection performance and based on the results of this campaign more wells were stimulated in 2024. After these successful stimulations, injectivity is not an issue and as of November 2024 an injection

rate of above 26 kbbls/d has been realized without any problems. This has been a marked improvement as previously, injectivity could not exceed 10 kbbls/d.

Field-F also has a relatively high degree of heterogeneity, and this has been addressed by the successful LEL installation in many wells. Conformance has significantly improved (as per the recent PLTs) and therefore LEL completions will be installed in some additional wells by the end of 2025.

Three key elements necessary for successful Field Development have been addressed, and what is currently pending is the determination of the optimal well spacing. The optimal well spacing together with the right injection and offtake rates are critical in ensuring that the water-drive concept works. This is essential to maximize the reservoir recovery factor.

Reservoir health is monitored by a comprehensive data acquisition plan. There is some depletion but nothing alarming at this stage. The VRR targets have been met this year despite quite a few operational challenges, and the plan is to achieve a CVRR around 1 in the coming years. With the sustained increase in injectivity, this is not an issue. The production targets are also comfortably met, and Field-F is part of production growth programs. A successful completion of the field development of such reservoirs would be a key enabler for the future Production Growth projects. Finally, since the current Recovery Factor (RF) is less than 1% it is difficult to assess the reservoir performance at such an early stage although managing the water-cut and GOR are extremely important.

Field-F is an integral part of the Digitalization program and the information of most of these wells can be monitored ensuring efficient reservoir management.

The successful implementation of key technologies could render Field-F as a Blueprint for analogue Field Development and Reservoir Management of tight reservoirs. This would help unlock Reserves and realize value. The learnings could be deployed in analogue reservoirs. Field-F is also a particularly good example of integration between multi-disciplinary teams collaborating to maximize value.

Field Overview

Field-F (Figure 1) is an under saturated tight reservoir with an average porosity of 21% and a permeability ranging from 0.01 to 3 mD. The field development concept is Waterflooding with a Line Drive pattern, and the oil producers are on Gas Lift. The production/injection performance from its early development scheme is constantly evaluated to further optimize the full field development strategy.

Field-F field development is taking place in distinct phases. The Field-F-PH-1 project consists of a total of 6 OP and 6 WI wells. For long term sustainability and to compensate for the production decline from other reservoirs, Field-F-PH-2 project is planned with additional 34 wells. Field-F will be de-risked by the dynamic performance of the initial development scheme. This will be further reevaluated through a set of planned activities.

Additionally, EOR production will be considered to increase production from this tight reservoir. Advanced EOR studies are planned to be conducted to find solutions to maximize the recovery factor.

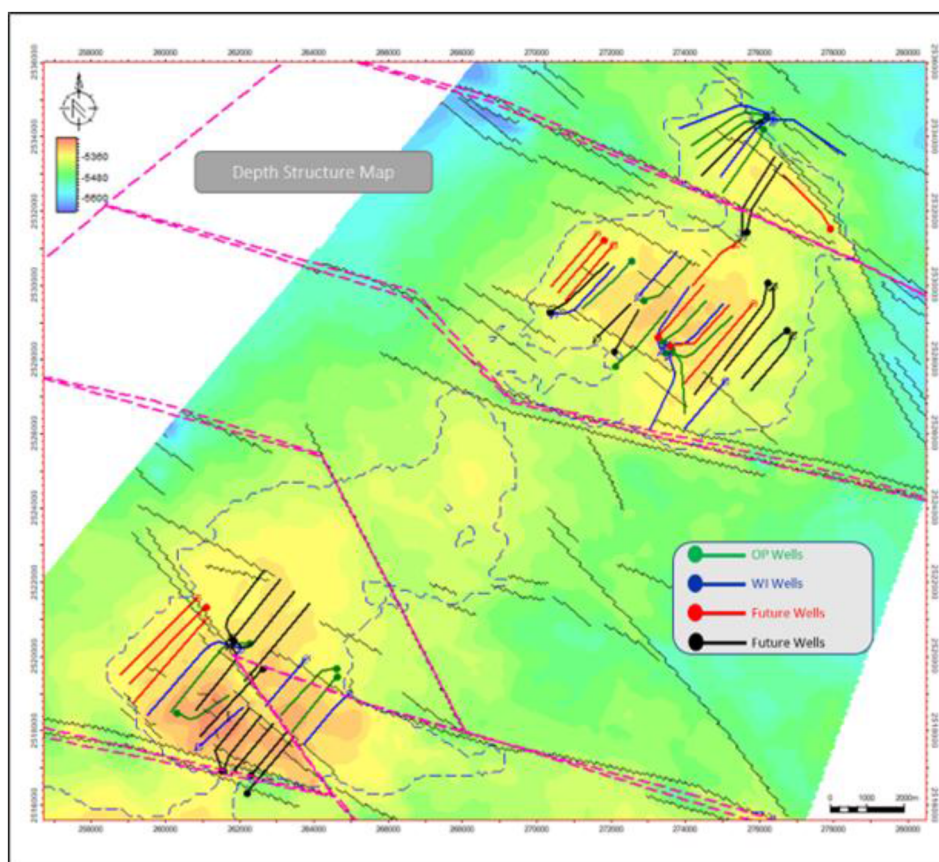


Figure 1—Field-F depth map.

Several technologies have been implemented since Field-F development began. These technologies have resulted in:

- Productivity Enhancement
- Injectivity Enhancement
- Conformance Improvement

In addition to the already applied technologies, the Field Development concept will be de-risked by:

- The implementation of a WAG pilot and
- The implementation of infill drilling (Down spacing from 500 m to 250 m).

Field-F reservoir health is monitored through comprehensive data collection, with well data easily accessible via FDIS as part of the Digitalization strategy.

In Field-F, not only were the right steps taken to develop the reservoir properly but also the implementation of these steps has been successful. Therefore, Field-F can serve as a blueprint for the successful development of analogue reservoirs, unlocking reserves and contributing to future growth plans.

Field Development Workflow

In this section, the various field development elements of tight reservoirs are listed and described briefly (Figure 2).

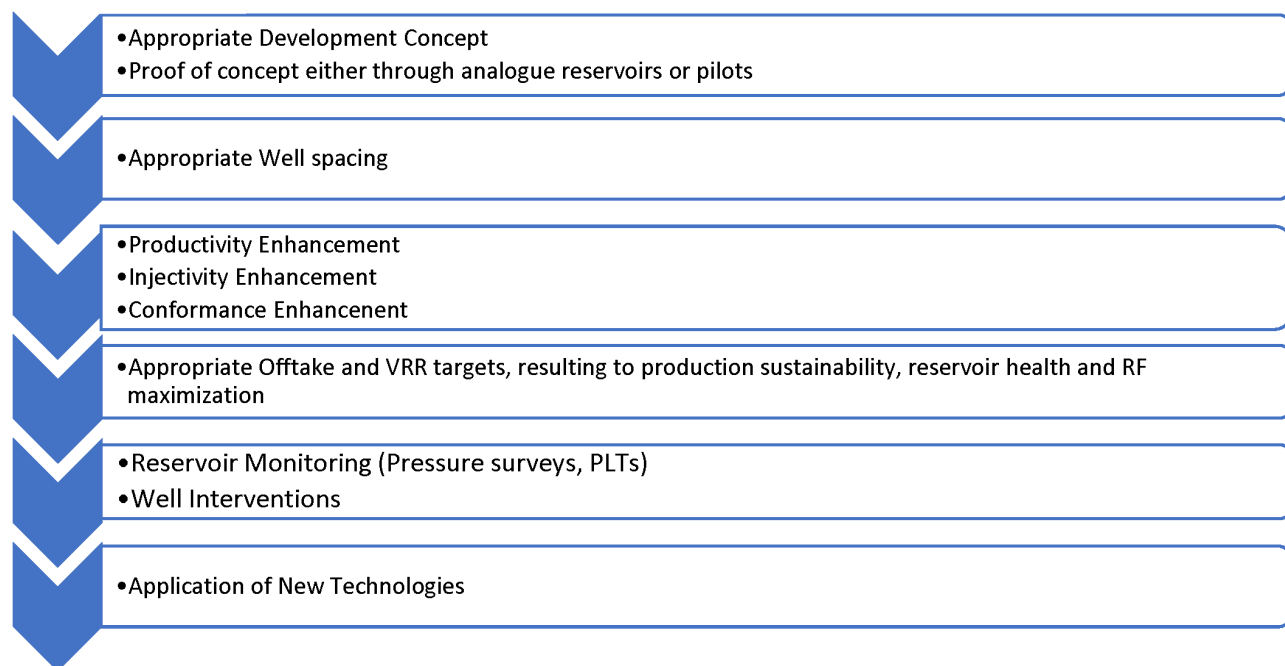


Figure 2—Schematic Summary of Tight Reservoir Development Process.

The first step of a successful field development is to select the appropriate development concept. For a tight reservoir, the most common development options are Waterflooding, Gas Injection or Water Alternating Gas Injection. The appropriate development concept depends on the fluid and rock properties, reservoir structure and architecture, oil distribution, and water and gas availability. Gas Injection and WAG development schemes are more prevalent in very tight reservoirs where injectivity issues can be serious. The proof of the development concept can be done either via pilots or via analogue reservoirs. The development concept will have a major impact on the Expected Ultimate Recovery and Project Economics.

Depending on the development concept, and the tightness and thickness of the reservoir, the appropriate well spacing is selected. It is especially important that the injectors are in communication with the corresponding producers, providing the appropriate pressure support. The distance will depend on the reservoir thickness and tightness and the type of injectant. Gas injection allows larger distance between injectors and producers while water injection requires tighter spacing. In case the producer and injector are not in communication, there will be an area of decreasing pressure around the producers and an area of increasing pressure around the injectors. In such cases, the water drive recovery mechanism breaks down and the Expected Ultimate Recovery suffers. The appropriate well spacing has a significant impact on the required number of wells, affecting UTC and Project Economics.

Once the development concept has been selected and the well spacing set, the productivity, injectivity and conformance issues need to be addressed. The productivity and injectivity issues are very pronounced in tight reservoirs and their successful treatment is the key to the field development success. Conformance enhancement is also key since it is linked to better sweep efficiency and recovery factor optimization. In Field-F productivity has been increased, using Hydraulic Fracking and Fishbone Stimulation. Injectivity has been increased with Advanced Acid Stimulation while conformance has been enhanced with LEL completions in the injectors.

The appropriate offtake and VRR targets need to be selected carefully to ensure production sustainability, maintain reservoir health and maximize recovery. Communication between injectors and producers is essential to ensure pressure maintenance and a controlled recovery from the reservoir. Throughput, as a rule of thumb, must be around 3% of the HCPV but also other factors need to be considered such as reservoir heterogeneity, production quotas and project economics.

A comprehensive Reservoir Management Plan is particularly important to collect the data required to manage the reservoir depletion efficiently and maximize recovery. The RMP consists of a number of BHCIPs, PFOs and PBU's, distributed across injectors and producers. PLTs are important to understand zonal contributions and the need of intervention in producers while in injectors they are used to understand and quantify the injection conformance. The RMP plan also provides the data necessary for a timely well intervention such as zonal isolations, water shut offs etc.

The last element is the use of New Technologies. Keeping up with the New Technology developments and understanding the applicability to our reservoirs is essential to optimize the field development and maximize recovery. This is especially important in tight reservoirs where recovery is more challenging than more permeable reservoirs.

New Technologies Selection

Field-F is a low permeability reservoir, and the main challenge is to find a way to improve the productivity and injectivity and increase the oil recovery. Field-F was evaluated considering different technologies as a pilot project (Figure 3). Fishbone Stimulation, Hydraulic Fracturing and long horizontal length wells have been implemented, and the results were very encouraging. The suitability of Fishbone Drilling has been assessed but because of the perceived complications and high cost it has not been piloted in Field-F.

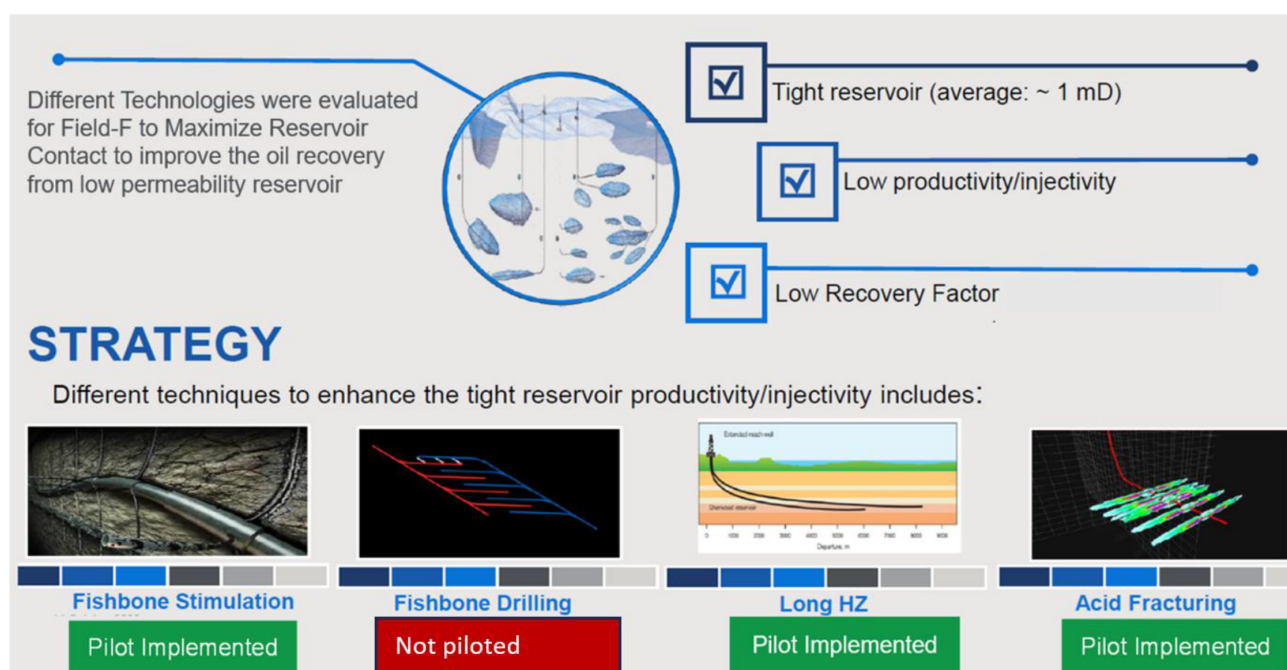


Figure 3—Field-F Technology Implementation.

Productivity Enhancement

The first step was productivity enhancement of all producers. So far, this has been done with Hydraulic Fracturing and Fishbone Stimulation. Both Hydraulic Fracturing and Fishbone Stimulation have been applied in one well each, with very encouraging results and sustained performance improvement.

Hydraulic Fracturing

Hydraulic Fracturing has been applied in well OP-HF, located in the Northern region of the reservoir. OP-HF is a horizontal well with a 6" open hole section with length of approximately 4800 ft. The reservoir is

limestone and shows a high content of calcite mineralogy. The permeability at the OP-HF location varies between 0.1 to 0.4 mD and the effective porosity reaches up to 25%.

The OP-HF Acid Frac job took two weeks to complete, and it consists of 12 stages in total. Each stage has two clusters. The perforation length is 1.5 ft per cluster (i.e., 3 ft perforation per stage) leading to a total perforation length of 36 ft. The total volume pumped during the acid frac job was around 20000 bbls (~1600 bbls per stage). After the end of the acid frac job, the well flowed back using Coiled Tubing and recovered approximately 3000 bbls of water with no oil. After the Flow to Clean (FTC) operation, the well started testing using PTS. The total fluid recovered during the FTC + testing was 31600 bbls.

Post Hydraulic Fracking, the well was tested using PTS (Figure 4) for a short duration to confirm the well potential and performance of Hydraulic Fracking. The well was tested at multiple choke sizes and the well performance showed encouraging results. OP-HF tested maximum up to 2800 bbls/d at 52/64" choke size with around 10% water cut. The produced water samples were analyzed and the salinity trend indicated that the well was still cleaning.

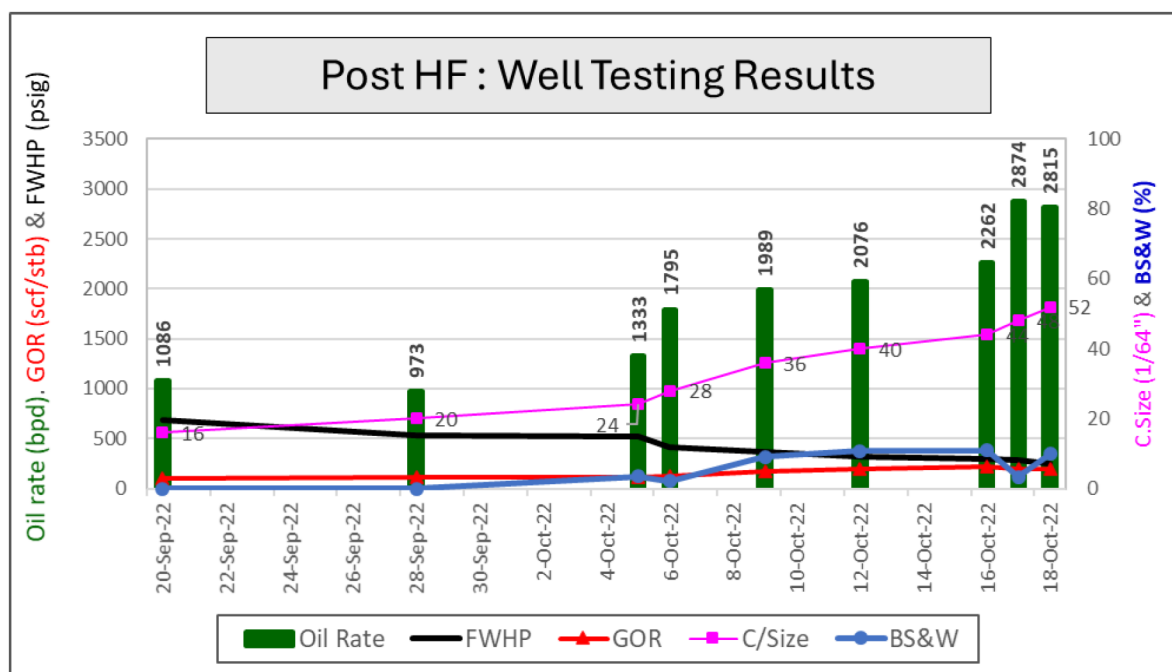


Figure 4—OP-HF Post-Acid Frac Well Testing results.

Post Hydraulic Fracking, a PLT job was conducted to examine the contribution from fracked stages and confirm the post-frac job well performance. The PLT indicated that three perforations (10685' to TD) were not contributing. That translates to an overall of 21% noncontributing perforation intervals.

Before the Acid Fracking, the well tested ~500 bbls/d or less with 23% water cut using PTS (Figure 5). Post-Acid Fracking the well tested, using PTS, ~2200 bbls/d with 14.5% water cut. The well produced at ~1800 bbls/d for the next three months. Post Hydraulic Fracking, OP-HF production performance improved significantly. However, post-workover (OP-HF was recompleted with a normal production string), it could not achieve that production level, which was most likely due to some damage during the workover resulting in a productivity decrease. Production performance is sustained until now without any loss of productivity and a moderate increase in water cut. The only drawback of the acid fracturing was the high cost.

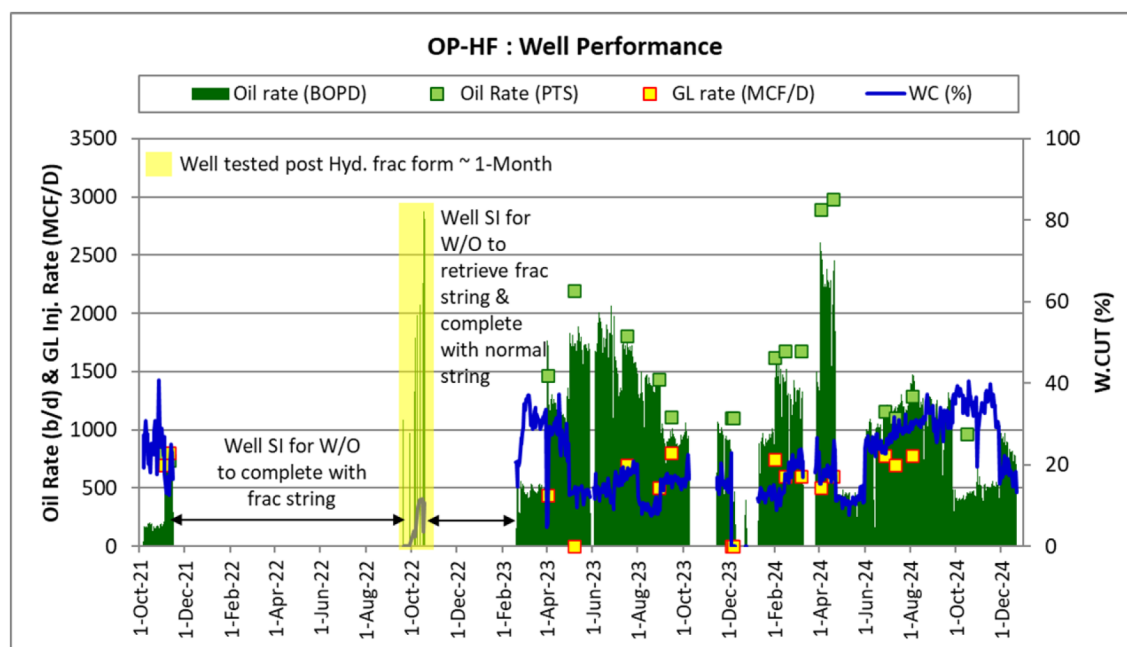


Figure 5—Pre- and Post-Acid Frac, OP-HF oil rate and water-cut.

Fishbone Stimulation

Fishbone Stimulation has been applied in well OP-FS, located in the Northern region of the reservoir as part of a production enhancement strategy from tight reservoirs. The OP-FS completion is shown in Figure 6 and it includes 40 fishbone subs. Each sub has four Fishbone Nozzles that are 40 ft long.

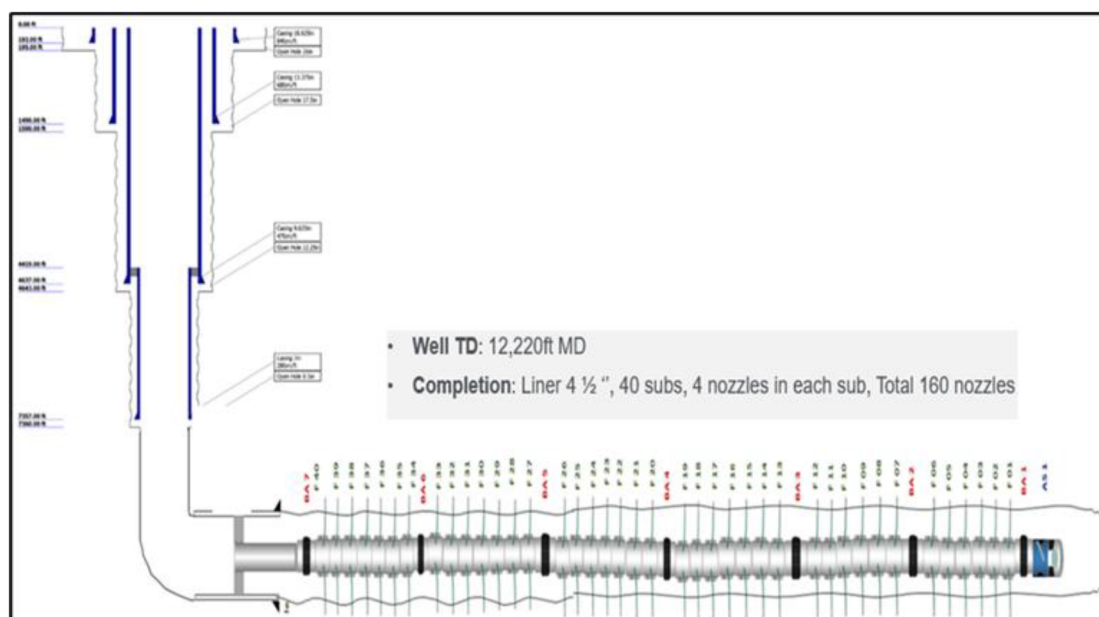


Figure 6—OP-FS well completion.

Well OP-FS started testing through PTS on October 26, 2021 (Figure 7). The well started production with a rate of 1870 bbls/d which gradually declined to around 645 bbls/d on natural flow after one week. During this period BS&W remained less than 10%. Gas lift started on November 2021, and the oil rate significantly improved to around 1000 bbls/d. PLT was conducted in December 2021 to determine the fluid contribution across the horizontal section. Based on PLT results, the well contribution was divided into three sections.

The PLT showed 74% of total contribution was from Section 1 and 26% of contribution was from Section 2. However, Section 3 was not contributing (there was stagnant water in the wellbore most likely due to well trajectory) as shown in Figure 8.

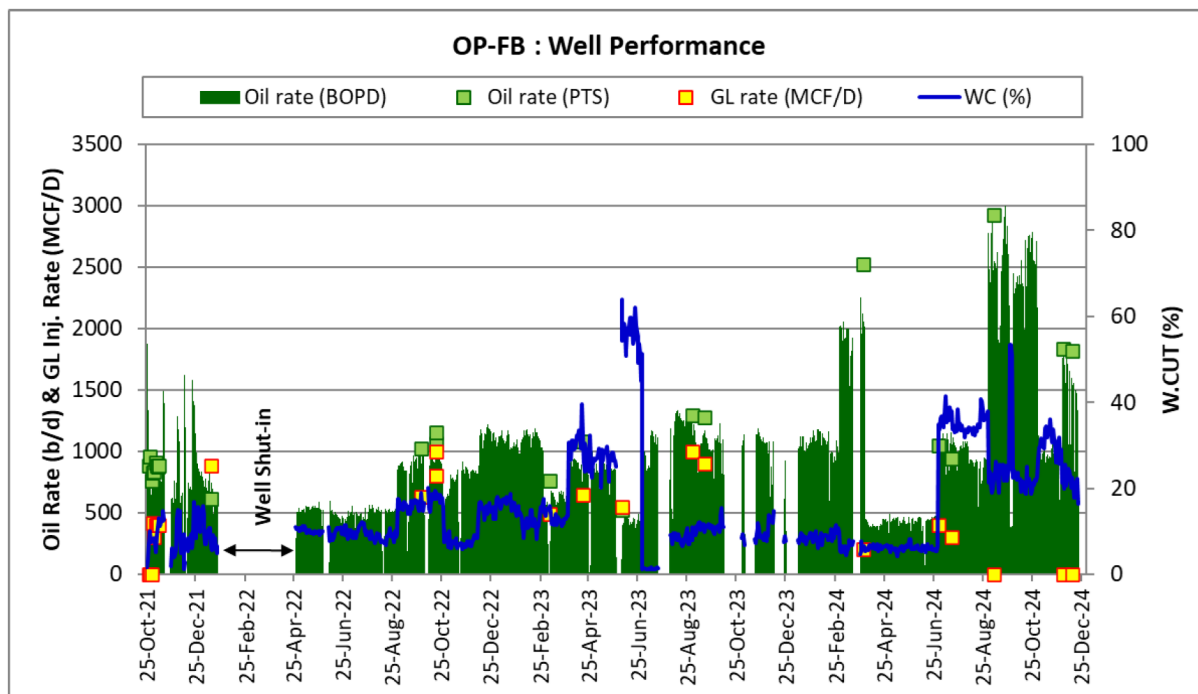


Figure 7—Post-Fishbone Stimulation Oil Rate and Water-Cut.

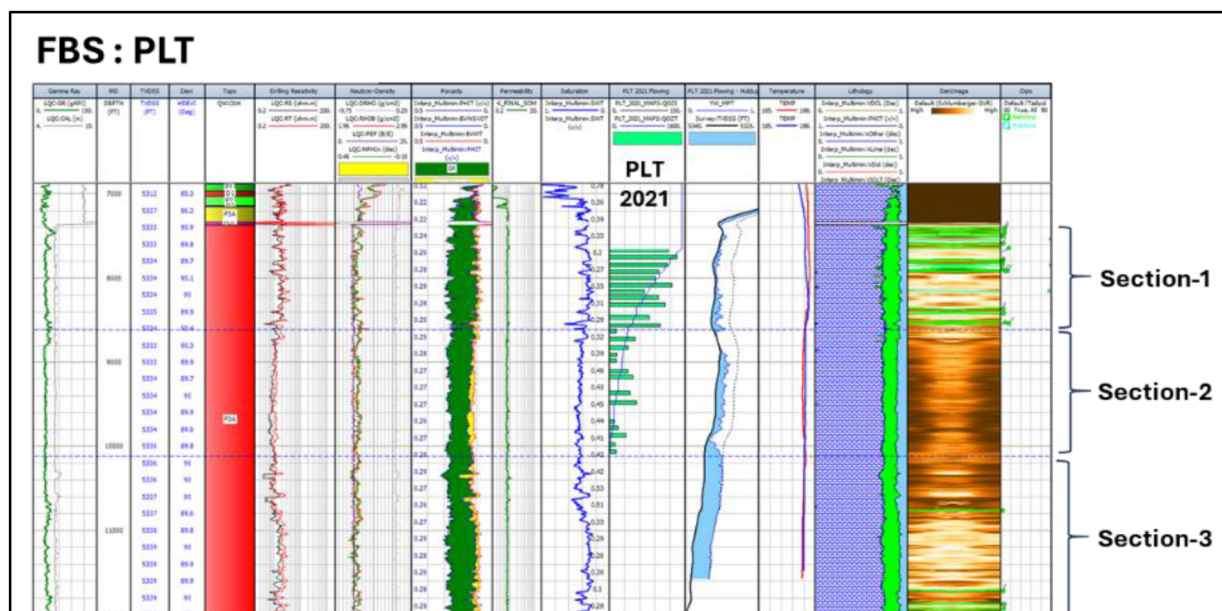


Figure 8—OP-FS PLT Analysis.

Oil rate gradually remained stable at 750 bbls/d for approximately two months and BS&W was approximately 7%. OP-FS produced approximately 1200 bbls/d from December 2022 to mid-February 2023. From August to September 2023, the well tested around 1400 bbls/d using PTS on GL. Currently the well is producing ~1100 bbls/d with 13% WC on Natural Flow due to reduced production quota. Production enhancement is sustained, and the well can produce up to 1900 bbls/d with a moderate WC.

Figure 9 shows Field-F production performance with and without Fishbone Stimulation and Hydraulic Fracturing. There is a significant production contribution from Fishbone Stimulation and Hydraulic Fracturing, amounting to approximately 29%.

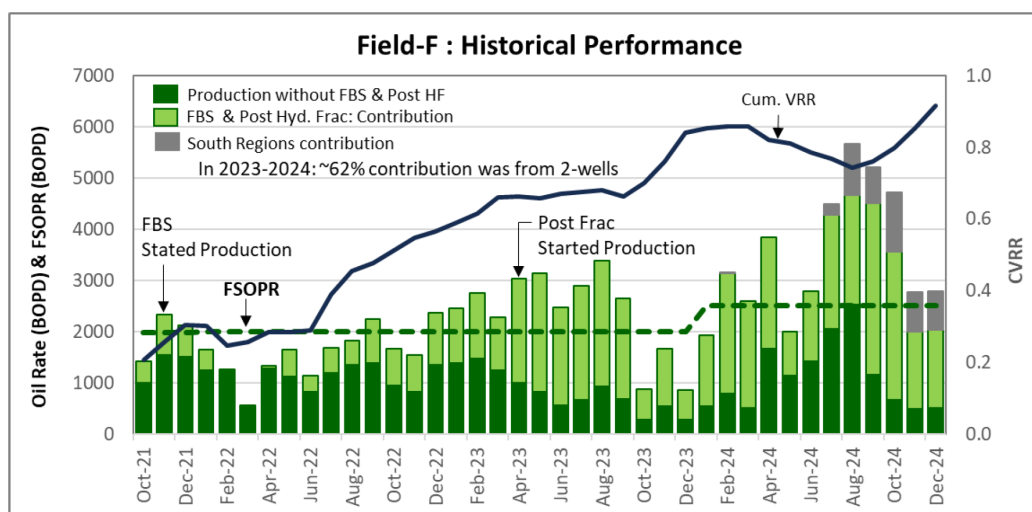


Figure 9—Field-F performance with and without Hydraulic Fracturing and Fishbone Stimulation.

In Figure 10 one can compare the performance of the Fishbone Stimulated well OP-FS to the offset wells (OP-A, OP-B and OP-C). All four wells are drilled in the same area; they are Open Hole completions and have similar lengths. The only difference between them is the Fishbone Stimulation. The production performance is markedly different while the water-cut is remarkably similar. As with the hydraulically fractured well (OP-FS), productivity enhancement is sustained until now.

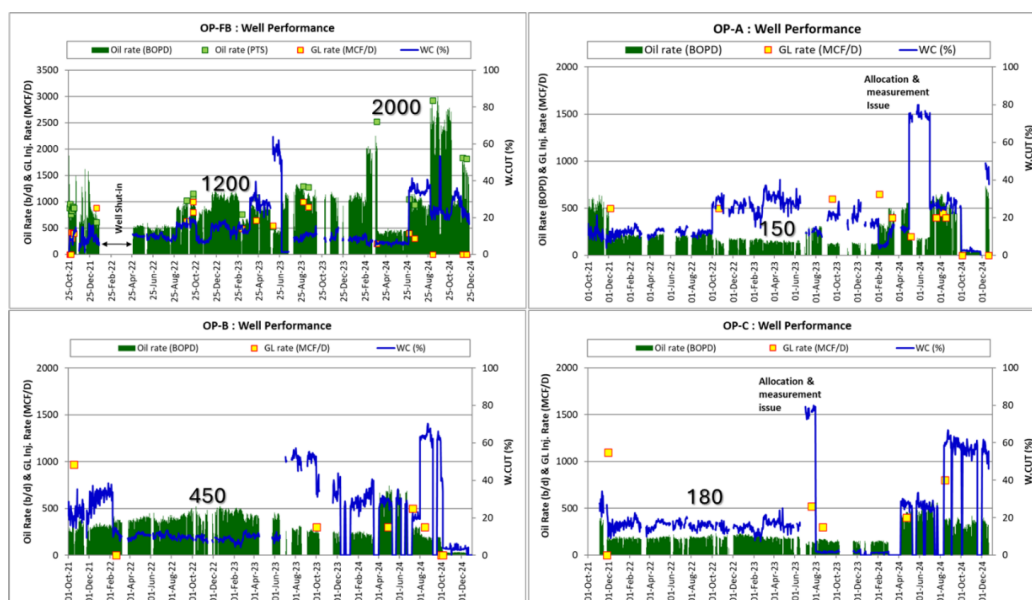


Figure 10—Oil Rate and Water-Cut comparison of OP-FS and its offset wells.

In Figure 11, the sustained rates of the Hydraulically Fractured (OP-HF) and Fishbone Stimulated (OP-FS) wells are compared with the conventionally treated wells in the same reservoir. The sustained rate improvement is three- to five-fold.

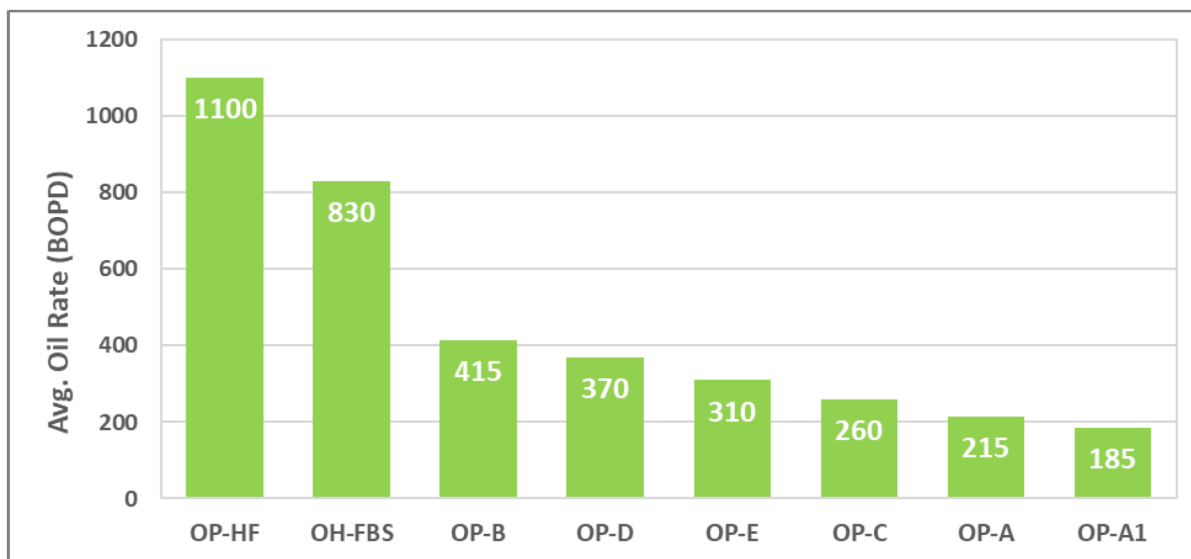


Figure 11—Comparison of the average oil rate for the Hydraulically Fracked, Fishbone Stimulated and the "normal" offset wells.

Injectivity Enhancement

The next step of the successful field development was to enhance injectivity. This has been achieved with Emulsified Acid Stimulation. The Emulsified Acid improved worm-holing due to its lower diffusivity compared to HCl and increased viscosity, providing better diversion.

Advanced Acid Stimulation

This was the first-time application of Emulsified Acid Abrasive Jetting. Agitator tools were used to improve wellbore accessibility and there was a Water Injection cluster ESP upgrade to increase pressure and volume delivery by an additional 10 kbbls/d. High velocity jetting tools were used for maximum worm-holing in the reservoir. As a result, the Voidage Replacement Ratio (VRR) has increased significantly (Figure 12) and the Cumulative VRR target has been consistently met.

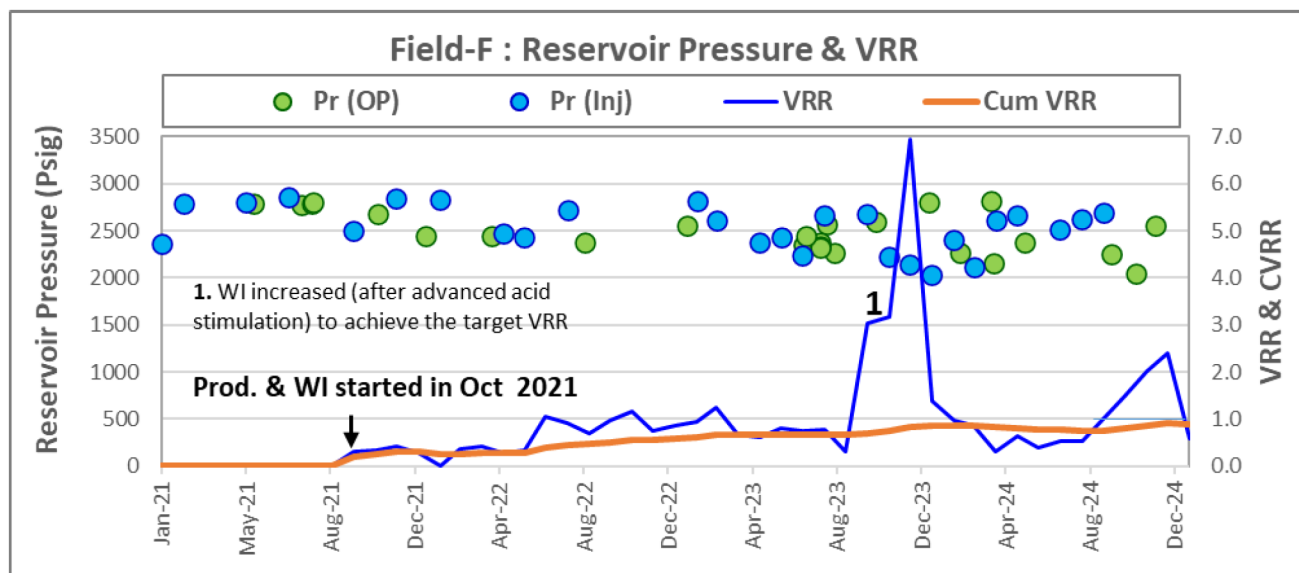


Figure 12—Field-F Pre- and Post-Acid Stimulation VRR and CRR.

In Figure 13 one can see the pre- and post-stimulation water injection performance of the first four water injectors that were subjected to Advanced Acid Stimulation. For WI-A the water injection rate has improved three-fold however the FWHIP was still high. For WI- B the water injection rate has increased 3.8 times while the WHIP has reduced by 60% (from 1378 psig to 522 psig). WI-C has seen the highest water injection increase by 4.5 times, but the FWHIP remained high at 1650 psig. Finally, WI-D has seen the most modest water injection increase (~2.5 times) but with a reduction in FWHIP.

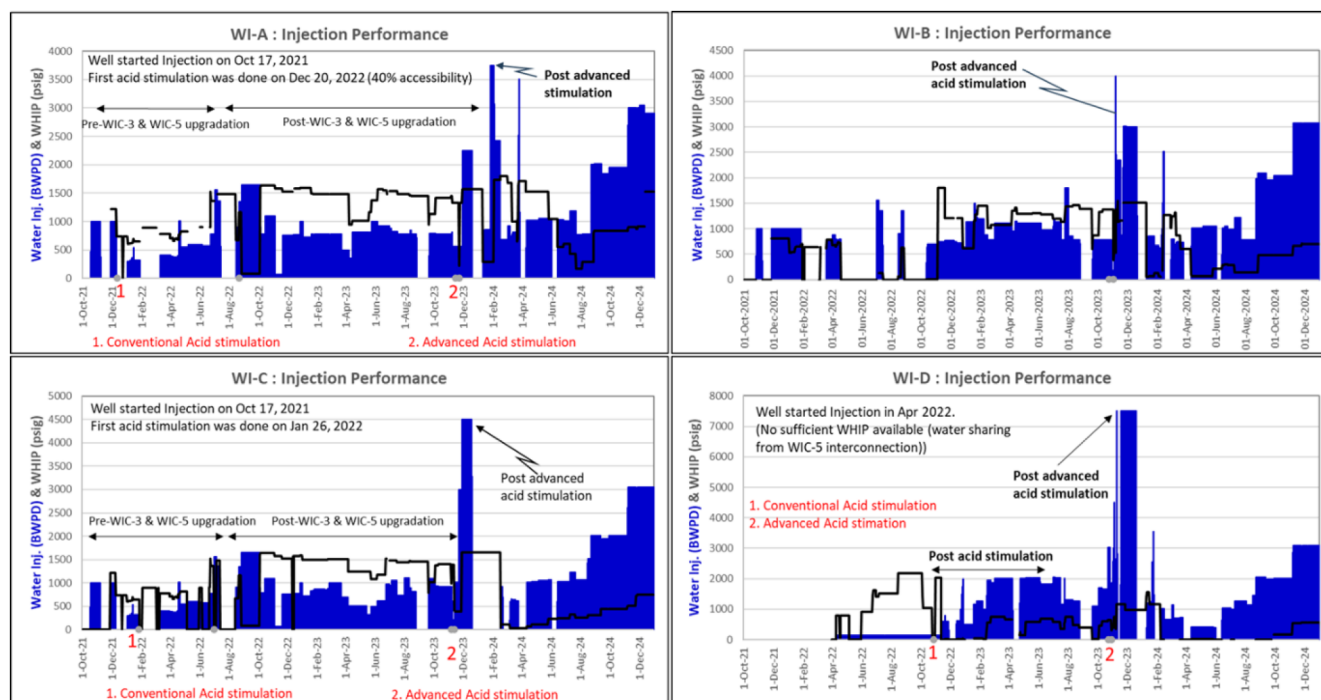


Figure 13—Pre- and Post-Stimulation water injection performance for four injectors.

Currently all water injectors have been stimulated, and the combined water injection rate easily exceeds 25 kbbls/d, covering all water injection needs, maintaining the reservoir pressure, and helping achieve the VRR targets.

Additional De-Risking Activities

Apart from the productivity, injectivity and conformance enhancements the following development activities are planned to de-risk and further optimize the field development:

- Two WAG pilots (4 wells, 2 OP + 2 WAG) will be implemented in 2025 followed by long term performance evaluation. Field- F is a good candidate for gas injection and a successful GI/WAG pilot will further optimize the Field Development, increase the Recovery Factor and improve the project Economics.
- Advanced Stimulation will be piloted in some Oil Producers.
- Additional wells are scheduled to undergo lower completion (LEL).
- Phase-2 wells (Infill drilling) - Evaluation of optimal well spacing. This step is key for the optimal Field Development and project Economics optimization.
- Maximum Reservoir Contact (MRC) wells evaluation is ongoing. A successful evaluation of MRC wells will further optimize the field development and result in a reduction of the required number of wells.

Table 1—Added value and comparison.

	Average Rate (bbls/d)	Incremental Cost	Incremental Value
Conventional	267	-	-
OP- FS	1015	25%	419%
OP-HF	1041	82%	403%

Conclusions

Field-F, a tight reservoir, has seen significant improvements in productivity, injectivity and conformance through the successful implementation of various technologies. The reservoir is at initial stages of development, but the production targets are easily met. VRR targets are met and the Cumulative VRR is on target to reach 1. A proper de-risking strategy is in place and many additional improvement initiatives are planned to optimize the field development and further improve the project Economics. The improvement initiatives and the technologies employed are varied and multi-disciplinary and showcase the complexity and challenges of developing such tight reservoirs. Nevertheless, these technologies and initiatives have been planned, deployed, and executed successfully in Field-F making it a blueprint for the successful development of analogue reservoirs, unlocking reserves, and contributing to future growth plans.

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